Female Labor Participation and Gender Gap in Latin America (1990–2025)

1.Introduction

This project analyzes the evolution of **female labor force participation** and the **gender participation gap** across selected Latin American countries between **1990 and 2025**.

The analysis was conducted using **Power BI**, based on data from the **World Bank – World Development Indicators**. The objective is to identify long-term trends, regional patterns, and differences across countries in labor market participation by gender.

This project was developed as part of a **data analytics learning portfolio**, focusing on the application of **data visualization, data modeling, and analytical metrics using DAX in Power BI**.

2. Data Source

Source:
World Bank – World Development Indicators

Indicators used:

- Female Labor Force Participation (% of population ages 15+)
- Male Labor Force Participation (% of population ages 15+)
- Female Unemployment Rate
- Male Unemployment Rate

Countries included in the analysis:

- Argentina
- Brazil
- Chile
- Colombia
- Mexico
- Peru
- Uruguay
- Venezuela

3. Methodology

The analysis was conducted using **Power BI** following these steps:

1. Data Preparation

- Data cleaning and formatting
- Integration of male and female labor participation datasets
- Standardization of country and year variables

2. Analytical Metrics

Several measures were created using **DAX**, including:

Gender Gap

Male Labor Force Participation − Female Labor Force Participation

This measure represents the difference between male and female labor participation rates.

Additional calculated measures included:

- Gender Gap in **1990**
- Gender Gap in **2025**
- **Change in Gender Gap (1990–2025)**

3. Data Visualization

A Power BI dashboard was developed to present the results through:

- Summary KPI indicators
- Time-series trend analysis
- Cross-country comparisons
- Analytical tables
- Insight summaries

4. Key Findings

4.1 Increase in Female Labor Participation

The analysis shows that **female labor force participation has increased across most of the countries analyzed since 1990**.

The trends indicate a gradual increase in female participation over the past three decades, with some temporary fluctuations around **2020**, likely associated with the labor market disruptions caused by the COVID-19 pandemic.

These findings are consistent with broader regional studies highlighting the increasing integration of women into the labor market in Latin America.

4.2 Regional Reduction in the Gender Gap

The results indicate that **the gender participation gap has decreased in most countries over time**.

Between **1990 and 2025**, the regional gap declined by approximately **9.6 percentage points**, suggesting gradual progress toward greater gender balance in labor market participation.

Although the reduction is not uniform across all countries, the overall trend suggests a regional convergence in participation rates between men and women.

4.3 Significant Differences Across Countries

The comparative analysis also reveals **substantial differences between countries**.

For example:

- **Peru records the highest female labor force participation in 2025**, reaching approximately **65%**.
- Other countries show lower participation levels, reflecting different economic structures and labor market dynamics.

These differences highlight the importance of national economic and institutional contexts in shaping labor market outcomes.

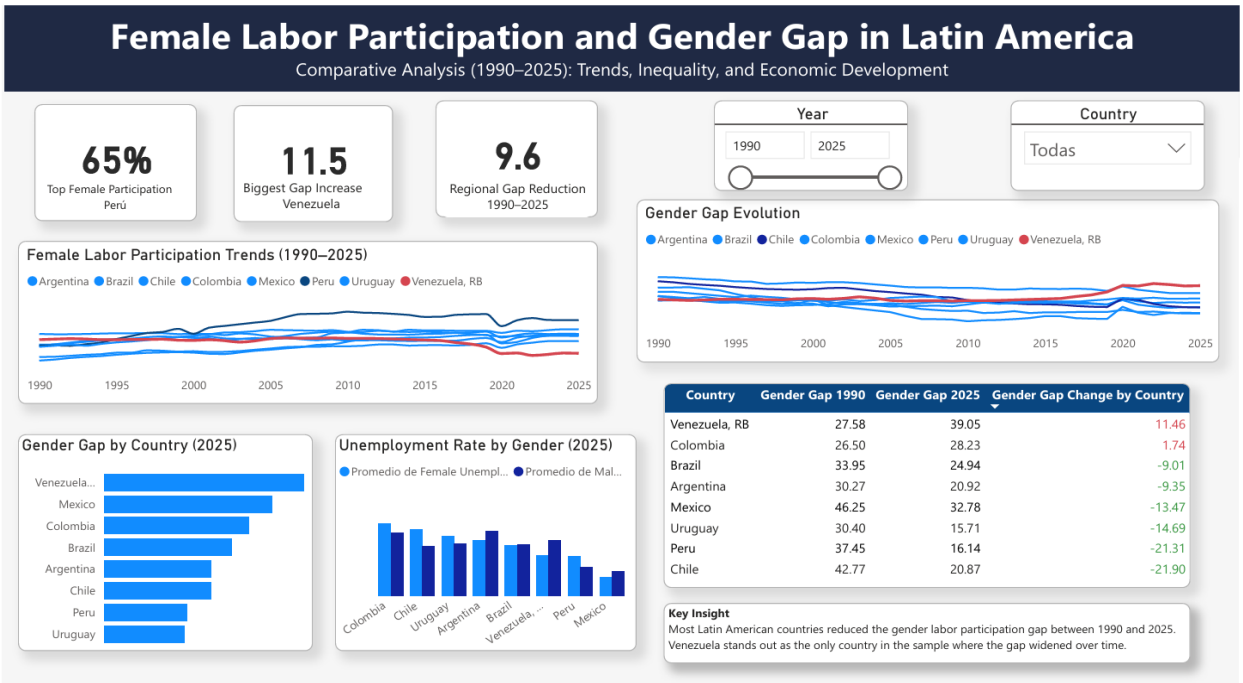
4.4 An Outlier Case: Venezuela

One of the most notable findings of the analysis is the case of **Venezuela**.

While most countries experienced a **reduction in the gender participation gap**, Venezuela is the only country in the sample where **the gender gap widened between 1990 and 2025**.

The analysis shows an increase of approximately **11.5 percentage points** in the participation gap, contrasting sharply with the regional trend.

This result suggests that economic and institutional conditions may significantly affect gender participation dynamics in labor markets.



Country	Gender Gap 1990	Gender Gap 2025	Gender Gap Change by Country
Venezuela, RB	27.58	39.05	11.46
Colombia	26.50	28.23	1.74
Brazil	33.95	24.94	-9.01
Argentina	30.27	20.92	-9.35
Mexico	46.25	32.78	-13.47
Uruguay	30.40	15.71	-14.69
Peru	37.45	16.14	-21.31
Chile	42.77	20.87	-21.90

5. Dashboard Insights

Based on the visual analysis and calculated metrics, several insights emerge:

- Female labor participation has shown a **steady upward trend across Latin America over the past three decades**.
- The **gender participation gap has narrowed in most countries**.
- There are **important differences between countries**, reflecting diverse labor market structures.
- **Venezuela stands out as an exception**, where the gender gap increased instead of decreasing.

6. Conclusions

The analysis suggests that Latin America has made **gradual progress in reducing gender disparities in labor market participation**, mainly driven by increases in female participation rates.

However, the results also indicate that this progress **has not been uniform across countries**, highlighting the influence of structural, economic, and institutional factors.

The Venezuelan case illustrates how economic disruptions and institutional changes may significantly alter labor market dynamics and gender participation patterns.

Overall, while progress toward gender equality in labor participation has been made, **important differences remain across countries**, indicating that further improvements are still needed.

7. Tools and Technologies

- Power BI
- DAX
- Data Visualization
- World Bank Data

8. Project Value

This project allowed the application of several data analytics skills, including:

- data modeling
- development of analytical metrics using DAX
- comparative data visualization
- generation of insights from real-world datasets

and forms part of a **data analytics learning portfolio focused on economic data analysis and visualization**.